consumption, thermal management naturally gets simplified. Therefore, most processors aim to decrease the power consumption to take care of this combined problem.

Intel's approach for thermal management and power consumption is similar. Intel Cryo Cooling technology is one of the latest thermal management technologies for desktop and laptop processors. This technology takes advantage of overclocking to system stability by adjusting temperature and clocking frequency depending on the CPU workload. It automatically manages sub-ambient cooler power consumption according to the workload demands, and also helps avoid condensation risks. All these techniques are combined with sub-ambient CPU coolers with smart condensation control to offer lower condensation risk and practical power consumption.

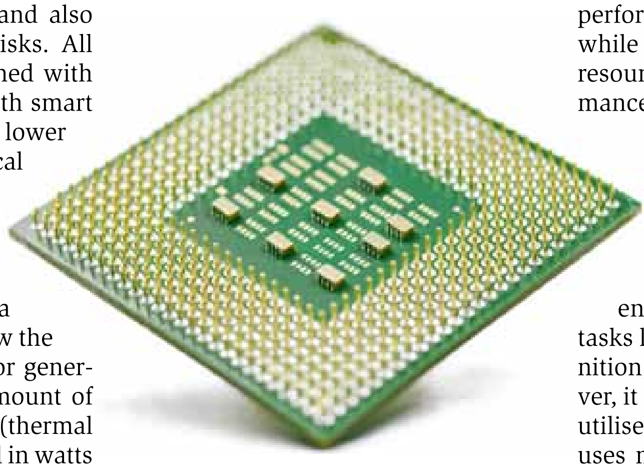
There are various thermal management systems available from AMD and Intel. But before making a choice, it is important to know the amount of heat the processor generates. The measure of the amount of heat generated is called TDP (thermal design power). It is measured in watts and refers to the power consumption under the maximum theoretical load.

The power consumption of the CPU will be less than TDP most of the time. However, this data of TDP that is being used to design the system is the power consumption when the system runs at maximum power. This ensures that the processor will be working reliably even under maximum workload.

The Intel Core i9-11980HK processor features configurable TDP with TDP-down of 45W and TDP-up of 65W. TDP-down mode is used when cooler and quieter operation is needed. This mode specifies lower TDP and lower frequency operation than the nominal value (processor's rated frequency and TDP). The TDP-up mode is used when extra cooling is available. Configurability here means that the computer manufacturer can modify the TDP of the CPU within the specifications, depending on the chassis and the type of cooler used.

Machine learning and deep learning capabilities

Today's intelligent computer systems can learn from the feedback taken to enhance user experience. So they can adjust operations after continuous exposure to data and other inputs. Machine learning (ML) refers to a process in which the machine actively learns for itself from various past inputs without being explicitly programmed. ML uses algorithms just like human neural networks, and these algorithms are designed to improve performance over time as they get exposed to more data.



The future performance can thus be enhanced by learning from the response received by the user.

Deep learning is a subset of machine learning that is more advanced and goes beyond the latter to solve problems. It makes use of deep neural networks to access, explore, and analyse vast sets of information. An easy example would be all the music files on Spotify being considered to make ongoing music suggestions based on the tastes of a specific user.

Latest i9 processors from Intel feature the Intel deep learning boost—a set of embedded processor technologies designed to accelerate AI deep learning use cases. It combines Intel Advanced Vector Extensions 512 (AVX-512) with a new vector neural network instruction (VNNI) that significantly increases deep learning inference performance over previous generations.

The AVX-512 is a new instruction set on the latest Intel CPUs that implements single instruction and multiple data (SIMD), which means that one CPU instruction is performed against multiple data items at the same time. Moreover, Intel Xeon processors embed SAP HANA machine learning capabilities. They run machine learning algorithms with minimum latency.

In situations where power and performance are necessary, Intel Gaussian & Neural Accelerator (Intel GNA) provides power efficiency always-on support. The GNA is designed to deliver AI speech and audio applications such as neural noise cancellation without the involvement of processing cores. Therefore, the GNA performs the AI speech-related tasks while simultaneously freeing up CPU resources for overall system performance and responsiveness.

The Apple M1 chip includes Apple neural engine for accelerating ML tasks. This engine consists of a 16-core architecture, capable of 11 trillion operations per second. This enhances the machine learning tasks like video analysis, voice recognition, and image processing. Moreover, it is more effective and efficiently utilises the hardware. Specifically, it uses machine learning to learn how to become more efficient in processing and power consumption based on historical data obtained from day-to-day user-to-device interaction.

The Mediatek's MT8192 and MT8195 Chrome book processors integrate a high-performance AI processing unit (APU) to power a wide range of voice and vision based applications. The APU combines the advantages of Edge AI and Mediatek's AI ecosystem, Neuropilto. This means that the AI processing is done on the device rather than on the cloud service, thus keeping the information about the device usage entirely private.

The Qualcomm 8cx Gen 2 5G processor comes with a dedicated Qualcomm AI engine, which means it uses the Snapdragon 8cx Gen 2 architecture at its full potential to deliver capabilities like intelligent camera, voice UI, and security, turning your PC into a hub of mobile productivity and entertainment. The AI engine uses deep learning methods for increased power efficiency across hardware and